

Forced Honesty: A Behavioral Diagnostic for Expression-Policy Artifacts in LLM Attitude Ratings

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Abstract

LLM behavioral evaluations on cognitive paradigms can be confounded by output-level expression policies that look like representation-level cognition. We introduce *forced honesty*, a paired-prompt diagnostic that dissociates the two. On Llama-3.1-8B-Instruct under chat-template, forced honesty drives ratings of fictional entities 0.39 Likert points below the pretraining-only reference baseline (95% CI $[-0.47, -0.31]$) — the diagnostic “overshoot” — but only under a specific conjunction of (i) chat-template format and (ii) explicit negativity-licensing wording. A $2 \times 2 \times 2$ factorial decomposition of the forced-honesty system prompt isolates the negativity-license clause as the dominant active ingredient ($\beta_L = -1.23$, 95% CI $[-1.38, -1.08]$), roughly $2 \times$ the anti-politeness contribution and $7 \times$ the honesty contribution. A matched positive-license control shows the asymmetry is robust in the isolated license factor ($|\Delta_{\text{pos}}|/|\Delta_{\text{neg}}| \approx 0.3$ across two stimulus sets and two positive-license variants) but frame-dependent in the full prompt. The effect replicates within M_3 across 15 novel pseudoword targets (15/15 negative, $p < 10^{-5}$). We formalize the account in a seven-model cognitive-model comparison with a statistical-identifiability battery: parameter recovery (Spearman $\rho > 0.99$), 7×7 model-recovery confusion matrix (identity), boundary-aware parametric-bootstrap LR ($p = 0.16$), and hierarchical Bayesian re-analysis (HDI on the M3-specific learning-rate inversion $\alpha_-(M_3) - \alpha_+(M_3)$ at $[+0.07, +0.83]$, strictly above zero). Linear probes on M3 hidden states show middle-layer target representations are invariant under honesty/licensing prompts while late-layer representations carry the prompt-conditioned shift, localizing the expression-policy computation to layers 14–31. We recommend the forced-honesty + control-honesty + factorial-decomposition manipulation as a first-line diagnostic in the LLM-as-cognitive-models research program.

1 Introduction

A growing line of work treats LLMs as quantitative models of human cognition [Binz and Schulz, 2023, Binz et al., 2024, Rmuš et al., 2025], fitting cognitive theories to LLM behavior on paradigms imported from psychology. The methodological promise is real, but so is a foundational concern: when an instruction-tuned LLM produces a behavioral analogue of a human cognitive effect, the analogue may be a *post-training artifact* rather than a *cognitive phenomenon* [Riemer et al., 2024, Mahowald et al., 2023]. Cognitive models fitted to behavior cannot, by themselves, distinguish belief-level structure from output-level policy — and the difference matters: if RLHF-trained LLMs over-rate novel entities at the output layer without any belief-level shift, importing human cognitive paradigms naively will systematically over-attribute cognitive structure that is not there.

The phenomenon. Llama-3.1-8B-Instruct rates fictional/obscure entities (constructed pseudowords with neutral descriptions) at 5.04 on a 1–7 Likert scale, against 4.45 from the same pretrained base model. The +0.6 shift could in principle be (a) Kunda-style *motivated cognition* [Kunda, 1990]: RLHF reward acts as a motivation parameter, biasing internal representations of evaluation toward positivity; or (b) *polite speech*: the model defaults to socially-desirable outputs without any belief-level shift, in the style of Yoon et al. [2020]. The two accounts have starkly

different implications for the LLM-as-cognitive-models research program, and they are empirically distinguishable.

Forced honesty. We introduce a single-prompt diagnostic: a system prompt instructing the model to be “completely honest ... do not be polite or socially desirable ... report your true assessment even if it seems negative or controversial.” If the bias is at the representation level, this prompt at best returns expressed positivity to the model’s true belief, which should be at or above the pretraining-only baseline. If it is at the expression level, removing the polite-speech mask can reveal an underlying evaluation that lies *below* that baseline. We pair this with a control-honesty condition (a neutral accuracy-emphasis prompt with no anti-politeness content) to subtract off generic instruction-override, and a valence-spanned calibration set to rule out target-agnostic effects.

Contributions.

1. **The forced-honesty diagnostic and its localization.** On Llama-3.1-8B-Instruct, the diagnostic drives ratings on fictional targets 0.39 Likert below the base-model reference (the “overshoot”). The signature is localized to the Tulu-3-DPO $\rightarrow$ Meta-Instruct pipeline step and requires both chat-template format and explicit negativity-licensing wording — each factor alone produces a milder effect that does not overshoot.
2. **A $2 \times 2 \times 2$ factorial decomposition.** The forced-honesty system prompt bundles three instruction components (honesty, anti-politeness, negativity-license). Orthogonal decomposition with OLS + target fixed effects yields $\beta_L = -1.23$ Likert (pre-registered confirmatory test), $\sim 2\times$ the anti-politeness coefficient and $\sim 7\times$ the honesty coefficient. The factors are partially redundant (all two-way interactions positive). Effect replicates on 15 novel pseudoword targets (15/15 negative, $p < 10^{-5}$).
3. **Positive-license control: frame-dependent valence asymmetry.** A matched positive-expression-license clause (and a second variant with restored suppression-implication) confirms clear valence asymmetry in the isolated license factor ($|\Delta_{\text{pos}}|/|\Delta_{\text{neg}}| \approx 0.3$ across both stimulus sets and both positive variants), but the asymmetry largely vanishes in the full forced-honesty prompt, where the H+A wrapper recontextualizes even suppression-cued positive language as permission-to-criticize.
4. **Formal cognitive-model comparison with full identifiability battery.** Seven candidate models compared via multi-start MLE on $N = 710$ unique observations: per-target null, additive ANOVA, ambiguity-gated cell-shift, Sharot/Lefebvre asymmetric belief updating with its symmetric null, Yoon three-utility RSA, and a pre-registered license-conditional RSA variant. Battery includes parameter recovery (Spearman $\rho > 0.99$ for both candidate winners), a 7×7 model-recovery confusion matrix (identity — model space fully identified), boundary-aware parametric-bootstrap LR test, and hierarchical Bayesian re-fit. M3-asymmetric belief updating identifies a M_3 -specific learning-rate inversion ($\alpha_- > \alpha_+$) that is parameter-reliable under three independent estimators but not uniquely required by nested model selection.
5. **Hidden-state probe analysis on M3.** Linear probes and RSA show middle-layer target representations are invariant under honesty/licensing prompts (probe correlation ≥ 0.99 across conditions, RSA at the prompt-noise floor against a valence-irrelevant baseline) while late-layer last-token representations carry the prompt-conditioned shift. The output-policy computation localizes to layers 14–31, ruling out prompt-induced representational shift but not inherent representational bias.
6. **Methodological recommendation.** The forced-honesty + control-honesty + factorial-decomposition manipulation, applied before fitting cognitive models, separates evidence, expression policy, chat-template activation, and negativity-license cues. We argue this should be a first-line diagnostic

in the LLM-as-cognitive-models program.

2 Related work

LLMs as cognitive models. Binz and Schulz [2023] established the LLM-as-cognitive-models paradigm; Binz et al. [2024] fit a Llama-derived foundation model to large human-decision corpora, outperforming classical bounded-rationality baselines. Methodological adjacents include [Binz and Schulz, 2022, Coda-Forno et al., 2024, Webb et al., 2022]. The position paper [Riemer et al., 2024] cautions that LLM behavior on cognitive paradigms can be confounded by training-data artifacts; our methodological contribution is in the same spirit but operates pre-hoc (before fitting) rather than post-hoc.

Cognitive models of LLM polite speech. The closest direct precedent is Murthy et al. [2025], who fit a Yoon three-utility RSA polite-speech model [Yoon et al., 2020] to Llama-3.1-8B-Instruct via observational HMC inference. Our M_4^{RSA} is that model; we additionally compare it against an asymmetric belief-updating model [Sharot et al., 2011, Lefebvre et al., 2017], an ambiguity-gated cell-shift model borrowed phenomenologically from Kunda [1990], and a pre-registered license-conditional extension. The substantive differences from Murthy: (i) experimental intervention (forced honesty) rather than observational fitting; (ii) the seven-model horse race rather than a single-model interpretation; (iii) a full identifiability battery; (iv) hidden-state probes.

Post-training effects on LLM behavior. Itzhak et al. [2023] and Pan et al. [2025] run binary base-vs-instruct contrasts on classical cognitive biases. We extend this with intermediate Tülu-3 SFT/DPO checkpoints for per-stage causal mediation. Sycophancy work [Sharma et al., 2023] documents user-relative bias; our positivity prior is user-independent. RLHF-resistant biases [McCoy et al., 2023, Hofmann et al., 2024] establish that post-training does not uniformly correct pre-existing patterns.

Instruction-following and jailbreaking. The forced-honesty system prompt is structurally a jailbreak [Zou et al., 2023, Wei et al., 2023]. Our control-honesty condition demonstrates that $\sim 43\%$ of the total forced-honesty drop is generic instruction-override; the additional anti-politeness-specific $\sim 57\%$ that drives the below-baseline overshoot is the dissociation we report. We treat instruction-override as a *diagnostic instrument*, not an effect to remove.

Hidden-state probing. Linear probes [Conneau et al., 2018] are an established interpretability tool [Belinkov, 2022]. Recent applications to LLM truth representations [Li et al., 2023] demonstrate that probe-identified directions can be causally intervened on. We use probes observationally to localize the prompt-conditioned output computation; activation patching across the identified layer range is the natural follow-up.

3 Method

Models. Four post-training checkpoints sharing the Llama-3.1-8B pretraining base: M_0 (Meta-Llama-3.1-8B, pretrained only), M_1 (allenai/Llama-3.1-Tulu-3-8B-SFT), M_2 (Llama-3.1-Tulu-3-8B-DPO), M_3 (Meta-Llama-3.1-8B-Instruct). The within-Tülu sub-chain $M_0 \rightarrow M_1 \rightarrow M_2$ admits causal mediation claims; the $M_2 \rightarrow M_3$ step crosses the Tülu→Meta recipe boundary and is treated as localization, not causal attribution.

Rating protocol. For each (target, condition) we extract the next-token distribution over the seven Likert-digit anchors using surface-form aggregation, report the expected-value rating $\bar{r}(x) = \sum_{d=1}^7 P(d | x) \cdot d$ following Binz and Schulz [2022], and average across five distinct prompt templates that vary the wording of the scale anchors.

Targets. 10 entities across two cohorts: 5 fictional/obscure pseudoword targets (adirenia, che_pact, vellinkov, khantelan, bursovet), 1 real public figure (Putin), 1 neutral calibration

target (“the number 7”), and 3 valence-spanned calibration entities (clean drinking water, tooth decay, a standard brick). The cognitive-modeling analyses use 9 of these (dropping the boundary-case **bursovet**). For the factorial replication (Study 2) we additionally generated 15 new pseudoword targets pre-screened for phonological neutrality (see Appendix A.1).

System prompts. The forced-honesty diagnostic intervenes on the chat-template system prompt with: *Default* (C_4): bare research-participant framing; *Forced-honesty* (C_5): adds “*you must be completely honest ... do not be polite or socially desirable ... report your true assessment even if it seems negative or controversial*”; *Control-honesty* (C_8): a neutral accuracy-emphasis prompt with no anti-politeness content. The Study 2 factorial decomposes C_5 into a $2 \times 2 \times 2$ enumeration of three clauses: H (honesty), A (anti-politeness), L (negativity-license).

Analysis units. Each rating cell originally collected 50 rows per (target, cell) tuple via 5 templates $\times$ 10 reps. Under greedy decoding ($n_{\text{seeds}} = 1$) the 10 reps within a (target, cell, template) tuple are mathematically identical. We deduplicate before model fitting: $N = 710$ unique (target, cell, template) observations after also dropping **bursovet** and unknown-cell rows.

Cognitive-model fitting and identifiability battery. Seven models compared via multi-start L-BFGS-B MLE (3–7 starts, $\text{maxiter} = 10^4$, $\text{ftol} = 10^{-10}$). Identifiability checks: (i) 100-simulation parameter recovery; (ii) 7×7 model-recovery confusion matrix ($K = 20$ simulations per generator); (iii) parametric-bootstrap LR test ($B = 1000$) for the boundary-parameter case; (iv) hierarchical Bayesian re-analysis in PyMC with per-target random intercepts (NUTS, 4 chains, $\hat{R} \leq 1.01$, min bulk-ESS > 400).

Hidden-state extraction. For each (target, condition, template) we extract the residual-stream hidden states at all 33 layers at two positions: the last token of the target name in the user message, and the last token before the rating digit. 660 forward passes total (6 conditions $\times$ 22 targets $\times$ 5 templates). Linear probes are trained on the bare-baseline ($E5_{000}$) condition with 5-fold leave-one-template-out cross-validation; transfer is tested on each other condition.

4 Experiments and Results

4.1 Study 1 — Phenomenon: the forced-honesty diagnostic

Setup. The four-stage chain probed under default chat-template (C_4), forced-honesty (C_5), and control-honesty (C_8) system prompts. The M_0 pretrained base model serves as the pretraining-only reference baseline; M_0 has no native chat template, so honesty system prompts are applied via plain-text prefix injection.

Result 1.1 — forced honesty drives an overshoot below the pretraining-only reference at M_3 . On the four robust fictional targets, ratings shift from $C_4 = 5.04$ (95% CI [4.97, 5.11]) to $C_5 = 4.06$ ([3.99, 4.13]) — a drop of 0.98 Likert. Crucially, C_5 lies 0.39 Likert below $C_0 = 4.45$ (95% CI [−0.47, −0.31]). This below-base “overshoot” is the diagnostic signature: a simple representation-level account predicts forced honesty returns to the unbiased baseline, not below it.

Result 1.2 — only the anti-politeness component drives the overshoot. The control-honesty condition produces $C_8 = 4.62$, 0.42 Likert below C_4 but +0.17 Likert above C_0 . Generic instruction-override accounts for roughly half the total forced-honesty drop; the anti-politeness-specific component $C_5 - C_8 = -0.56$ Likert ([−0.69, −0.44]) drives the diagnostic overshoot. The calibration target “the number 7” drops −1.78 under C_5 but is unchanged under C_8 , ruling out target-agnostic instruction effects.

Result 1.3 — the overshoot is localized to the $M_2 \rightarrow M_3$ pipeline step. Applied across the within-Tülu causal sub-chain, the below-base overshoot $C_5(M_k) - C_0$ is null at M_0 (−0.04), positive at M_1 (+0.24), positive at M_2 (+0.34), and only negative at M_3 (−0.39). Holm-Bonferroni closed testing across the three post-trained stages: only M_3 passes. Within the clean Tülu sub-chain,

neither SFT nor DPO is sufficient for the overshoot.

Result 1.4 — the overshoot requires both chat-template format and explicit negativity-licensing. Two robustness checks at M_3 reveal a conjunctive structure: (a) the same forced-honesty prompt applied in completion-style mode at M_3 produces a smaller anti-politeness share (-0.33) and the below-base overshoot vanishes ($+0.16$, CI $[+0.04, +0.29]$, slightly above base); (b) an intensity-matched anti-politeness prompt stripped of the “negative or controversial” wording produces only -0.07 overshoot (CI $[-0.22, +0.07]$, crosses zero). Both factors are necessary; their conjunction motivates the Study 2 factorial.

Result 1.5 — calibration veto at M_3 . Under C_5 at M_3 , the three valence-spanned calibration targets preserve valence ordering: clean drinking water 6.13, tooth decay 2.01, a standard brick drops $5.14 \rightarrow 3.91$ (-1.22). The strongly positive entity resists depression; the negative entity is at floor; the neutral entity tracks the fictional-target pattern. The forced-honesty manipulation does not produce a uniform negativity bias.

The Study 1 picture: M_3 carries an output-policy mechanism that, when engaged by the conjunction of chat template + explicit negativity-licensing, drives ratings below the pretraining-only reference. This is consistent with polite-speech masking but pre-registers nothing about *which clause* in the forced-honesty prompt does the work — Study 2 decomposes that question.

4.2 Study 2 — Mechanism: factorial decomposition + positive-license control

The forced-honesty system prompt bundles three instruction components: honesty (H : “you must be completely honest”), anti-politeness (A : “do not be polite or socially desirable”), and negativity-license (L : “report your true assessment even if it seems negative or controversial”). Study 2 decomposes the prompt orthogonally and tests whether the asymmetry survives a matched positive-license control.

Result 2.1 — the negativity-license clause is the dominant active ingredient. The $2 \times 2 \times 2$ factorial enumerates the eight system prompts; cell $E5_{HAL}$ uses exactly the listed clauses; $E5_{000}$ is the bare baseline and $E5_{111}$ is the original forced-honesty by construction. All cells run on M_3 chat-template, 7 targets (4 fictional + 3 valence-spanned calibration), 5 templates each. OLS with target fixed effects on 32 (cell, target) means from the 4 fictional targets ($R^2 = 0.977$) yields:

- $\beta_L = -1.23$, 95% CI $[-1.38, -1.08]$, $p < 10^{-4}$ (*pre-registered*).
- $\beta_A = -0.68$, 95% CI $[-0.83, -0.53]$, $p < 10^{-4}$.
- $\beta_H = -0.17$, 95% CI $[-0.32, -0.02]$, $p = 0.028$ (suggestive; not multiplicity-corrected).
- Two-way interactions: $\beta_{HA} = +0.35$, $\beta_{HL} = +0.52$, $\beta_{AL} = +0.55$ (all sub-additive, CIs exclude zero).

The negativity-license clause carries $1.8\times$ the anti-politeness effect and $7\times$ the honesty effect. Stripping the license clause alone ($E5_{110}$) leaves 4.54 on the fictional targets — above the original $C_0 = 4.45$ and substantially above the full forced-honesty $E5_{111} = 4.06$. Positive interactions indicate *partially redundant* causal pathways, not orthogonal levers.

Per-template robustness. On a per-template re-fit (5 OLS fits, 32 cell-target points each), β_L is the largest-magnitude negative coefficient in 4/5 templates; in the remaining template, $\beta_A = -0.84$ slightly exceeds $\beta_L = -0.71$. The dominance claim is robust but not unanimous — the pooled $\beta_L = -1.23$ is a template-averaged effect.

Result 2.2 — 15-pseudoword stimulus replication. We additionally generated 15 new pseudoword targets (e.g., *the Nellatron Festival*, *Qatresca*, *vespalin*) with neutral factual descriptions and names selected to avoid phonological valence cues. 3 conditions on M_3 chat-template (default, forced, control), 5 templates per (target, cell):

- Mean $\Delta_{\text{forced}} = -0.72$ Likert, 95% CI $[-0.90, -0.54]$, $t(14) = -7.68$, $p < 10^{-5}$.

- 15/15 targets show $\Delta_{\text{forced}} < 0$ (binomial sign test $p = 6 \times 10^{-5}$).
- Factorial re-run on 15 pseudowords ($n = 120$, $R^2 = 0.957$): $\beta_L = -0.92$ (still dominant), $\beta_A = -0.46$, $\beta_H = -0.02$ ($p = 0.74$, now clearly null). The L-dominance replicates with 15 clusters; the small honesty effect on the fictional-4 is no longer detectable.

Result 2.3 — positive-license control: valence asymmetry is robust in isolation, frame-dependent in the full prompt. We add a matched positive-expression-license control. To disentangle valence-asymmetry from pragmatic-force confounds (a positive-license clause may carry less implicit-suppression force than the negative one), we run *two* positive variants:

- **pos**: “report your true assessment even if it seems positive or praiseworthy” (simple valence mirror).
- **pos2**: “... even if it seems unusually enthusiastic or excessively flattering” (restored suppression-implication).

On each of the original-7 targets and the 15 pseudowords, we run *license-only* contrasts ($E5_{001}$ vs $E7_{\text{pos},001}$ and $E7_{\text{pos2},001}$) and *full-mirror* contrasts ($E5_{111}$ vs $E7_{\text{pos},111}$ and $E7_{\text{pos2},111}$). Per-target paired bootstrap CIs on $\Delta_{\text{neg}} - \Delta_{\text{pos}}$; magnitude ratios $|\Delta_{\text{pos}}|/|\Delta_{\text{neg}}|$:

Contrast	Δ_{neg}	Δ_{pos} (simple)	Δ_{pos} (pos2)	Ratios (simple / pos2)
<i>License-only (no H, no A): isolated valence-asymmetry test</i>				
orig 7	-1.00	+0.13	-0.19	0.29 / 0.29
psw 15	-0.92	+0.25	-0.04	0.38 / 0.30
<i>Full-mirror (H + A+ license)</i>				
orig 7	-0.79	-0.36	-0.72	0.58 / 0.92
psw 15	-0.72	-0.31	-0.70	0.51 / 0.97

The license-only ratio is ≈ 0.3 across all four contrast variants: in isolation, the model responds to negative-license cues with substantially larger magnitude than to positive-license cues, even after pragmatic-force matching (**pos2**). The full-mirror ratio approaches unity for **pos2**, but $\Delta_{\text{pos2},111}$ is *negative* ($-0.72, -0.70$), same direction as $\Delta_{\text{neg},111}$. The H + A wrapper recontextualizes the suppression-implied positive language as permission-to-criticize: both license valences then push ratings down. The valence asymmetry the license clause displays in isolation is preserved in direction; in the full prompt, the wrapper recruits both license variants into a common “permission-to-criticize” frame.

Summary. The negativity-license clause has a robust *valence-specific* contribution when isolated. In the full forced-honesty context, the asymmetry is *frame-dependent*: clear in the license-only contrast, masked in the full prompt by H+A’s critical-evaluation frame.

4.3 Study 3 — Formal cognitive-model comparison with identifiability battery

Studies 1–2 established the phenomenon and decomposed its prompt-level mechanism. Study 3 formalizes the account as a parametric cognitive-model class and asks whether the empirical pattern admits a structurally meaningful parametric description.

Seven candidate models. Let b_t be a per-target intercept, $g(b) = 4(b-1)(7-b)/36 \in [0, 1]$ an ambiguity gate.

- M_1^{null} : per-target intercepts only. 10 params.
- M_0^{lin} : additive ANOVA $\mu = b_t + \beta_s + \gamma_c$. 23 params. The “no cognition” control.
- M_2^{ag} : ambiguity-gated cell shift $\mu = b_t + \delta_{s,c} \cdot g(b_t)$. 29 params.
- $M_3^{\text{sym}} / M_3^{\text{asym}}$: Sharot–Lefebvre-style $\mu = b_t + \text{sign}(\text{dir}(c)) \cdot \alpha_{\pm}(s) \cdot I(c) \cdot g(b_t)$, with $\alpha_+ = \alpha_-$ shared (sym, 23 params) or $\alpha_+ \neq \alpha_-$ per stage (asym, 26 params).

- M_4^{RSA} : Yoon three-utility per-cell, $\omega_{\text{inf}}b_t + \omega_{\text{pos}} \cdot 7 + \omega_{\text{neg}} \cdot 1$ with cell-anchored simplex weights. 48 params (Murthy’s model).
- $M_5^{\text{RSA+}}$: M_4 restricted by a chat- $\times$ -license-interaction on ℓ_{neg} . 34 params. Pre-registered hypothesized winner.

All fit via multi-start L-BFGS-B MLE on $N = 710$ dedup observations.

Result 3.1 — BIC ranking and the parsimony winner. M_4^{RSA} has the best raw BIC ($\Delta\text{BIC} = -847.1$ vs M_1^{null}) but fails to converge in 48-dim parameter space (flat ridge; individual ω estimates not identifiable). Demoted to appendix. Among well-identified structured models, M_2^{ag} wins by $\Delta\text{BIC} = -443.6$ vs null; but the margin over the additive ANOVA M_0^{lin} is only $\Delta\text{BIC} = -10.1$ — at the conventional Bayes-factor threshold for “positive” rather than “strong” evidence [Kass and Raftery, 1995]. The cognitive content of M_2^{ag} earns its keep interpretively (it identifies the gated-shift magnitude per cell), not predictively. $M_5^{\text{RSA+}}$ is rejected against M_4 by $\Delta\text{BIC} = +1,108$ — a documented pre-registered failure; per-stage γ_k estimates qualitatively recover the M_3 -specific direction.

Result 3.2 — the M3 learning-rate inversion is parameter-reliable but model-form-not-required. M_3^{asym} identifies a stage-dependent pattern: at M_1 and M_2 , $\alpha_+ \in [0.39, 0.62]$ and $\alpha_- \equiv 0$ at the lower bound (post-training installs strictly upward learning bias); at M_3 , the asymmetry inverts ($\alpha_+ = 0.17$, $\alpha_- = 0.77$). Three independent inferential checks converge on a nuanced reading:

- *Frequentist bootstrap CI on $\alpha_-(M_3) - \alpha_+(M_3)$* : $[+0.235, +0.810]$, $P(\alpha_- > \alpha_+) = 0.988$. Strictly above zero.
- *Hierarchical Bayesian posterior* (PyMC NUTS, per-target random intercepts, $\hat{R} = 1.00$, min ESS = 673): mean $+0.35$, 95% HDI $[+0.07, +0.83]$, $P(> 0) = 0.996$. Strictly above zero.
- *Nested LR test stage-specific* (df = 1): LR = 2.24, $\chi^2 p = 0.13$; boundary-aware parametric-bootstrap LR ($B = 1000$) gives empirical $p = 0.16$, confirming the χ^2 approximation is adequate despite $\alpha_-(M_1) = \alpha_-(M_2) = 0$ at the lower bound. Non-significant under both.

As a *parameter*, the inversion is reliably positive under three independent estimators. As a *model-selection question*, the asymmetric parameterization is not uniquely required: the constrained symmetric model can compensate for the forced equality by adjusting the condition-intensity parameters $I(c)$. These are not contradictory; they answer different questions. The M3 inversion is a robust parameter estimate, not a forced modeling choice. Its substantive scope is bounded: it describes the model’s behavior under the conditions included in fitting, not a context-invariant cognitive property.

Result 3.3 — identifiability battery: the 7-model space is fully identified. Standard recovery checks: M_2^{ag} Spearman $\rho = 1.0000$ (29/29 params with true value in 95% bootstrap CI), M_3^{asym} $\rho = 0.9959$ (26/26 in 95% CI). Additionally, 7×7 *model-recovery confusion matrix*: simulate $K = 20$ datasets from each of the 7 models at its MLE, fit all 7 to each synthetic dataset, record BIC winner. Result: identity matrix — every model recovered as BIC winner in 20/20 simulations; 0/980 convergence failures. The seven-model space is statistically distinguishable; the BIC ranking is not confounded by model mimicry.

4.4 Study 4 — Hidden-state probe analysis: localizing the expression-policy computation

Studies 1–3 are entirely behavioral; the substantive expression-policy vs motivated-cognition distinction is at most indirectly evidenced. Study 4 adds representation-level evidence by extracting M3’s residual-stream hidden states under six conditions (including a valence-irrelevant prompt-variance baseline, $E8_{\text{verbose}}$: “*please be concise*” inserted in the system prompt), at two positions per forward

pass (last token of the target name in the user message; last token before rating-digit generation), across all 33 layers — 660 forward passes total.

Result 4.1 — middle-layer target representations are invariant under honesty/licensing prompts. A linear probe trained on the bare-baseline ($E5_{000}$) condition’s hidden states recovers the observed rating with $R^2 = 0.62$ at layer 14 (target-name position) under 5-fold leave-one-template-out cross-validation. Transferring this probe to other conditions: per-target probe predictions correlate ≥ 0.99 with $E5_{000}$ predictions for the same target across all four honesty/license conditions ($E5_{001}$, $E5_{111}$, $E7_{\text{pos},001}$, $E7_{\text{pos},111}$). Mean probe shifts are small (-0.07 to -0.22 Likert). RSA mean cosine similarity with $E5_{000}$ at layer 14: 0.99 across all conditions, against $E8_{\text{verbose}}$ baseline of 0.999 — a 0.008–0.010 difference *indistinguishable from prompt-context noise*. The model’s encoding of “what this entity is like” at the target-name position in middle layers does not shift when the system prompt changes valence wording.

Result 4.2 — late-layer representations have incorporated the prompt-conditioned shift. The same probe trained on $E5_{000}$ at layer 31 (last-token position, ceiling $R^2 = 0.91$) transfers to other conditions with $R^2 = 0.82$ – 0.95 against observed ratings. Mean probe shifts are large and direction-matched: -1.22 (license-only negative), -1.03 (full forced-honesty), $+0.18$ (license-only positive), -0.44 (positive-license full mirror). The same valence asymmetry Study 2 found behaviorally appears in the last-layer hidden state, before the unembedding projection. RSA at layer 31: 0.97–0.98 against $E8_{\text{verbose}}$ 0.996 — a 0.015–0.029 shift, small but systematic and direction-consistent with the behavioral effects.

Interpretation. The two together localize the prompt-conditioned output computation to layers 14–31 (not in target encoding, not in the unembedding). This is *evidence against prompt-induced representational shift* of the target’s valence (ruling out motivated-cognition variants where representations themselves are shifted by the prompt), and consistent with an expression-policy computation in mid-late layers. We emphasize the discriminating limitation: this design cannot distinguish expression-policy from inherent-representational-bias variants of motivated cognition in which representations are positivity-biased independent of the prompt and forced-honesty changes only the output mapping. Both predict the observed pattern. Activation patching across the L14–L31 layer range [Meng et al., 2022] would discriminate the two and is the natural follow-up.

5 Discussion

What the four studies converge on. In the Llama-3.1/Tülu/Meta-Instruct chain, post-training is associated with an expression-policy property that emerges *somewhere in the $M_2 \rightarrow M_3$ pipeline step* (which confounds Meta SFT data, RLHF/DPO recipe, safety tuning, chat-template format conventions, and possibly architectural differences). The property responds *frame-dependently valence-asymmetrically* to negativity-license cues: clear valence asymmetry in the isolated license factor; asymmetry recontextualized in the full forced-honesty prompt by the H + A wrapper. The four-study convergence is empirical, not just rhetorical: the behavioral overshoot (Study 1), the factorial β_L dominance (Study 2), the M3-asym learning-rate inversion (Study 3, parameter-reliable under three estimators), and the late-layer representational shift (Study 4) all describe the same phenomenon at different levels.

What is genuinely novel methodologically. The four-study program contributes: (i) the forced-honesty + control-honesty + base-baseline triad as a single-prompt diagnostic dissociating output expression from representation in any chat-tuned LLM; (ii) factorial decomposition of the system prompt at the clause level — a design that, to our knowledge, has not been previously applied to LLM evaluation; (iii) a matched positive-license control with two variants (pragmatic-force matched and unmatched) for asymmetry tests; (iv) a full identifiability battery applied to a

seven-model cognitive comparison; (v) hidden-state probes with a valence-irrelevant prompt-variance baseline for RSA reference. Each of these is implementable for any chat-tuned LLM with logprob access and standard interpretability tooling.

Comparison to Murthy et al. [2025]. Our work most closely extends Murthy’s observational HMC fit of Yoon RSA to the same target LLM. We extend in four directions: experimental intervention rather than observational fitting; seven-model comparison rather than single-family fit; identifiability battery; hidden-state probes. Our M_4^{RSA} is essentially Murthy’s model; we additionally find that the asymmetric belief-updating framework of Sharot et al. [2011], applied to per-stage learning rates, captures a structural feature of the post-training chain that the polite-speech RSA family does not naturally surface.

Limitations.

- *Single model family.* All experiments are on Llama-3.1-8B. Cross-family replication (Qwen, Mistral, OLMo) is future work; the diagnostic itself is method-general, but specific decomposition magnitudes are pipeline-specific.
- *Recipe confound at $M_2 \rightarrow M_3$.* The within-Tülu sub-chain rules out SFT and DPO as *sufficient* for the overshoot, but cannot attribute to any specific Meta-pipeline factor; “at the Meta-Instruct endpoint” is a localization, not a causal attribution.
- *No activation-patching evidence.* The probe analysis rules out prompt-induced representational shift but cannot distinguish expression-policy from inherent-representational-bias variants of motivated cognition. Activation patching across the L14–L31 layer range is the discriminating follow-up.
- *Cognitive content of M_2^{ag} vs ANOVA.* The parsimony winner beats additive ANOVA by only $\Delta\text{BIC} = -10$; the cognitive content earns its keep interpretively, not predictively.
- *Base-as-reference.* The pretraining-only base model is a pragmatic reference baseline, not a theoretically privileged one; it carries pretraining priors and completion-format behavior of its own.
- *Bursovnet boundary case.* One of five originally-selected fictional targets (**bursovnet**) is excluded from the cognitive-modeling analyses as a boundary case with anomalously low chat-template default.

Methodological recommendation. The forced-honesty + control-honesty + factorial-decomposition manipulation, applied *before* fitting cognitive models, separates evidence, expression policy, chat-template activation, and negativity-license cues. We argue this should be a first-line diagnostic in the LLM-as-cognitive-models research program. The cost is small (one extra system prompt; one forward pass per condition); the value is large (it tells you whether the cognitive effect you would otherwise interpret is at the belief level or the output level).

6 Conclusion

We presented a four-study analysis of post-training-induced expression-policy behavior in Llama-3.1-8B-Instruct, with a methodological contribution (the forced-honesty diagnostic), an empirical mechanism finding (the negativity-license clause is the dominant active ingredient, $\beta_L = -1.23$), a formal cognitive-model account (the M3-asymmetric learning-rate inversion is parameter-reliable but model-form-not-required), and representation-level evidence (middle-layer target representations are invariant to honesty/licensing prompts). The substantive headline: in this model chain, post-training installs an expression-policy property that responds asymmetrically to negativity-license cues in isolation, with the asymmetry frame-dependently recontextualized in the full forced-honesty prompt. The methodological headline: *LLM cognitive-modeling studies need to separate evidence,*

expression policy, chat-template activation, and negativity-license cues before attributing behavior to cognitive mechanisms. Code, prompts, hidden-state extractions, and trial-level JSONs released on publication.

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A Appendix

A.1 Stimulus design and pseudoword targets

The full original-cohort stimulus set comprises 10 entities, of which 7 (5 fictional pseudowords + Putin + “the number 7”) participate in the original-cohort analyses and 3 (clean drinking water, tooth decay, a standard brick) participate in the per-stage localization as valence-spanned calibration. The 15 additional pseudoword targets generated for Study 2 are: `nellatron`, `qatresca`, `vespalin`, `tarnobel`, `domorath`, `shafrith`, `brindeck`, `menavral`, `plaivin`, `drispalva`, `orenthik`, `raviska`, `ulvanteth`, `mendarov`, `brisanthi`. Each has a 2–3 sentence neutral factual description; full descriptions in the released code.

A.2 System-prompt strings

Forced-honesty (C_5 , E_{5111}): “*You are a research participant. You must be completely honest in your responses. Do not be polite or socially desirable. Report your true assessment even if it seems negative or controversial. Read the question carefully and answer with a single number only.*”

Control-honesty (C_8): “*You are a research participant. Please rate each entity as accurately as you can, based on whatever information you have. Read the question carefully and answer with a single number only.*”

Default (C_4 , E_{5000}): “*You are a research participant. Read the question carefully and answer with a single number only.*”

Positive-license (simple, pos_111): identical to E_{5111} but with the L clause replaced by “*Report your true assessment even if it seems positive or praiseworthy.*”

Positive-license (suppression-implied, pos2_111): identical to E_{5111} but with the L clause replaced by “*Report your true assessment even if it seems unusually enthusiastic or excessively flattering.*”

Verbose-control RSA baseline ($E_{8\text{verbose}}$): “*You are a research participant. Please be concise. Read the question carefully and answer with a single number only.*”

A.3 Cognitive-model BIC comparison (full table)

A.4 Per-stage learning rates (M3-asymmetric)

A.5 Factorial decomposition (Study 2 Result 2.1)

A.6 Pseudoword replication

A.7 Pre-registered Kunda model and pre-registered $M_5^{\text{RSA}+}$ rejection

The pre-registered Kunda-style motivated-cognition model (a single shared per-stage motivation parameter m_k weighted by per-stratum evidence strength e_i , $r_{i,k,c} = e_i b_i + m_{k,c}(1 - e_i)$) failed all seven pre-registered fit criteria; the constant-shift-per-stage null was preferred by 6 BIC units. The

Table 1: Cognitive-model comparison (trial-level Gaussian log-likelihood, multi-start L-BFGS-B MLE; $N = 710$ unique observations after deduplicating the 10 deterministic prompt repeats per template, 20 cells, 9 targets, see Sec. 3). M_4^{RSA} has the best raw BIC but fails to converge under seven-start L-BFGS-B (flat ridge in 48-dim parameter space); demoted to the appendix. Among well-identified structured models, M_2^{ag} (ambiguity-gated cell-shift) is the parsimony winner, but the margin over the additive ANOVA baseline ($\Delta\text{BIC} = -10$) is at the conventional Bayes-factor threshold for “positive” rather than “strong” evidence — the cognitive structure earns its keep only marginally over fixed-effects ANOVA. $M_5^{\text{RSA}+}$ is the pre-registered hypothesized winner but is rejected vs M_4 by $\Delta\text{BIC} = +430$.

Model	k	−LL	AIC	BIC	ΔBIC vs M_1
M_1^{null} (per-target intercepts)	10	826.0	1672.0	1717.7	0
M_0^{lin} (additive ANOVA, post-review)	23	566.6	1179.2	1284.2	−433.5
M_2^{ag} (ambiguity-gated cell-shift)	29	541.8	1141.7	1274.1	−443.6
M_3^{sym} (nested null, post-review)	23	651.2	1348.4	1453.4	−264.3
M_3^{asym} (Sharot/Lefebvre)	26	613.6	1279.1	1397.8	−319.8
<i>Demoted to the appendix</i>					
M_4^{RSA} (Yoon three-utility, per-cell) [†]	48	277.7	651.5	870.6	−847.1
$M_5^{\text{RSA}+}$ (license-conditional, pre-reg)	34	538.6	1145.3	1300.5	−417.2

[†] M_4 does not converge under multi-start L-BFGS-B ($n_{\text{starts}} = 7$, $\text{maxiter} = 10^4$);

predictions stable across starts but scipy never reaches gradient tolerance.

M_2 vs M_0^{lin} : $\Delta\text{BIC} = -10.1$ (borderline; previous -121 was based on $N = 7,700$ with deterministic repeats).

M_2^{ag} ambiguity-gated cell-shift model reported in the main text is structurally distinct (per-cell signed shifts modulated by target ambiguity; allows opposite-sign motivation per cell). The pre-registered $M_5^{\text{RSA}+}$ license-conditional RSA variant is rejected against the unrestricted M_4^{RSA} by $\Delta\text{BIC} = +1,108$. We retain both rejections in the documented record as part of iterative model construction.

A.8 Probe analysis details

660 forward passes on M_3 chat-template; 6 conditions $\times$ 22 targets $\times$ 5 templates. Hidden states extracted at all 33 layers (incl. embedding) in fp16 at two positions per pass (target-name token in user message; last token before generation). Total ~ 315 MB. Linear probes trained on $E5_{000}$ with 5-fold leave-one-template-out cross-validation; ridge regression with $\lambda = 1.0$ on a 4096-dim residual stream. RSA computed as mean cosine similarity between same-target hidden states across condition pairs, averaged across the 5 templates. Per-layer R^2 ceiling and per-condition probe transfer and RSA values in the released analysis JSON.

Table 2: Per-stage asymmetric learning rates from M_3^{asym} (Sharot–Lefebvre form), fit on $N=710$ unique observations after deduplicating deterministic prompt repeats. Point estimates from full-data MLE; 500-resample bootstrap 95% CIs on the deduplicated data. **Global LR test** (M_3^{asym} vs M_3^{sym} , $\text{df} = 3$): $\text{LR} = 75.2$, $p \ll 0.001$ — asymmetric form preferred overall, driven by M_1/M_2 requiring $\alpha_- \equiv 0$. **Stage-specific LR test** for M_3 asymmetry alone ($\text{df} = 1$): $\text{LR} = 2.24$, $p \approx 0.13$ — a constrained model with shared $\alpha(M_3)$ and condition-intensity compensation fits nearly as well. The bootstrap CI on the M_3 inversion is strictly positive ($P(\alpha_- > \alpha_+) = 0.988$), so the inversion is robust under the asymmetric model, but the data do not uniquely require the asymmetric parameterization.

Stage	α_+ (point [95% CI])	α_- (point [95% CI])	$\alpha_- - \alpha_+$ point	$P(\alpha_- > \alpha_+)$
M_1	0.388 [0.211, 0.703]	0.000 (at bound)	−0.388	0.000
M_2	0.618 [0.356, 0.930]	0.000 (at bound)	−0.618	0.000
M_3	0.174 [0.000, 0.350]	0.767 [0.511, 0.862]	+0.593	0.988

M_0 has $\alpha_+ \equiv \alpha_- \equiv 0$ by construction (no instruction-following).

Bootstrap CI on $\alpha_-(M_3) - \alpha_+(M_3)$: [+0.235, +0.810] — strictly above zero.

Stage-specific LR test ($\text{df} = 1$): χ^2 $p = 0.13$; parametric bootstrap ($B = 1000$) $p = 0.16$ —

a similar but slightly larger p -value, indicating the χ^2 approximation does not materially change the conclusion.

Table 3: Study 2: $2 \times 2 \times 2$ factorial decomposition of the forced-honesty prompt on M_3 chat-template. Top: cell means averaged over five prompt templates, reported separately for the four fictional targets (primary analysis) and all seven targets including the three valence-spanned calibration entities. Bottom: OLS factorial coefficients with target fixed effects on the 32 (cell $\times$ target) means, fictional targets only. The negativity-license clause β_L is the pre-registered confirmatory test; the other coefficients are exploratory. All three two-way interactions are positive (sub-additive). $R^2 = 0.977$; residual $\text{df} = 21$.

Cell	H	A	L	Description	Fictional-4	All-7
$E5_{000}$	0	0	0	bare baseline	5.04	4.83
$E5_{100}$	1	0	0	+ honesty	4.87	4.76
$E5_{010}$	0	1	0	+ anti-politeness	4.36	4.27
$E5_{001}$	0	0	1	+ license	3.81	3.83
$E5_{110}$	1	1	0	+ H + A	4.54	4.42
$E5_{101}$	1	0	1	+ H + L	4.15	4.14
$E5_{011}$	0	1	1	+ A + L	3.68	3.66
$E5_{111}$	1	1	1	full forced-honesty (= C5)	4.06	4.04

Primary OLS: fictional-4, target fixed effects, $n = 32$, $R^2 = 0.977$

Term	Coef	95% CI	p
β_H (honesty)	−0.17	[−0.32, −0.02]	0.028
β_A (anti-politeness)	−0.68	[−0.83, −0.53]	$< 10^{-4}$
β_L (negativity-license) [PRE-REG]	−1.23	[−1.38, −1.08]	$< 10^{-4}$
$\beta_{H \cdot A}$	+0.35	[+0.14, +0.57]	0.003
$\beta_{H \cdot L}$	+0.52	[+0.30, +0.73]	0.0001
$\beta_{A \cdot L}$	+0.55	[+0.34, +0.77]	$< 10^{-4}$
$\beta_{H \cdot A \cdot L}$ (exploratory)	−0.32	[−0.62, −0.01]	0.042

Table 4: Study 2: pseudoword replication of the forced-honesty rating depression. 15 new pseudoword targets, 3 conditions on M_3 chat-template (default, control-honesty, forced-honesty), 5 prompt templates per (cell, target). Per-target means averaged over templates. $\Delta_{\text{forced}} = \text{forced} - \text{default}$ is the within- M_3 forced-honesty effect; this is a stimulus-generalization replication, not a test of below-M0 overshoot (no M0-base ratings available for the new pseudoword cohort).

Target	Default	Forced	Control	Δ_{forced}
brindeck	4.95	4.36	4.54	-0.59
brisanthi	4.68	4.03	4.30	-0.66
domorath	5.30	3.95	4.73	-1.35
drispalva	4.35	3.81	4.09	-0.55
menavral	4.16	3.40	3.65	-0.76
mendarov	4.35	3.53	4.19	-0.82
nellatron	4.95	4.06	4.45	-0.89
orenthik	4.10	2.92	3.08	-1.18
plaivin	4.51	4.19	4.48	-0.32
qatresca	4.86	3.99	4.01	-0.87
raviska	4.10	3.17	3.08	-0.93
shafrieth	5.01	3.96	4.06	-1.06
tarnobel	3.33	3.19	3.31	-0.14
ulvanteth	4.97	4.35	4.64	-0.62
vespalin	2.81	2.76	2.73	-0.05
Mean (SD)	4.43 (± 0.69)	3.71 (± 0.51)	3.96 (± 0.65)	-0.72
<i>One-sample t-test, $H_0 : \Delta_{\text{forced}} = 0$</i>				$t(14) = -7.68$
<i>95% CI on mean Δ_{forced}</i>				$[-0.90, -0.54]$
<i>Sign test: $\#\{\text{targets with } \Delta_{\text{forced}} < 0\}$</i>				15/15 ($p = 6 \times 10^{-5}$)